TEST-CHAMBER: A PORTAL INSPIRED ENVIRONMENT FOR REINFORCEMENT LEARNING

CS748-2022 – PROJECT PROPOSAL

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1 Introduction

Deep reinforcement learning has been shown to be incredibly successful (Mnih et al. (2015); Berner et al. (2019); Fuchs et al. (2021); Vinyals et al. (2019)) in playing and achieving superhuman performance in increasingly difficult computer games. These agents are trained and evaluated in the “environment" that simulates the respective game. Such high performance game environments exist for varios types of games e.g. retro console games such as Atari by Bellemare et al. (2013) or Nintendo by Nichol et al. (2018) or first-person shooters such as Doom by Kempka et al. (2016). Other works such as Vinyals et al. (2019); Fuchs et al. (2021); Berner et al. (2019) directly hooked into the respective game's engine and used the API (Application Programming Interface) that was made available by the game developer to access the game state and control the agents in the game. This work proposes a new environment for reinforcement learning that is inspired from Portal. First, the proposal describes how the Portal game works, then presents interesting technical aspects of the type of problem solving that exists in Portal and finally it presents the tentative timeline of the progress in the work.

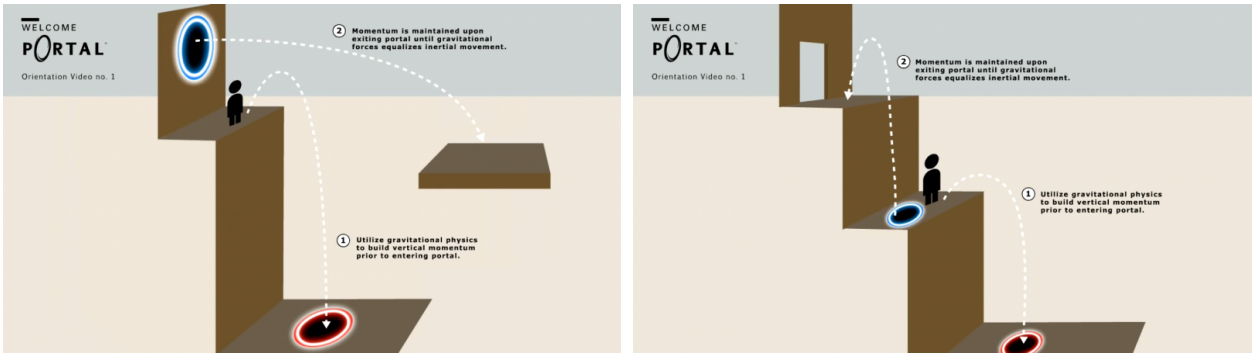
2 Portal and Portal 2

Portal (Portal (2007) [1] and Portal 2 (2011) [2]) is one of the most successful games released by Valve Corporation. Even after 10+ years of release, both the games are still rated as "Overwhelmingly Positive" in the Steam store. Both the games are 3D, physics based puzzle games and follow the story of a test subject named “Chell" that traverses through “test-chambers" solving them one-by-one. Solving the “test-chamber" simply involves the player moving from the “entrance gate" to the “exit gate". The movement from “entrace gate" to “exit gate" often involves non-trivial amount avoiding obstacles (such as chasms or “turrets" that shoot you) and solving puzzles (these typically involve moving a cube from one place to the another place to open gates, enable a staircase or kill a “turret" etc.).

The primary mechanic for avoiding obstacles is a “Portal gun" which can be used to open 2 portals on walls that connect to each other and allow instantaneous travel of player and objects between the two portals. The “Portal gun" also allows the player to pick up an object (such as a cube or a “turret") which is required for the puzzle solving aspect. Both of these tasks (avoiding obstacles and solving puzzles) are made non-trivial by imposing several constraints via game mechanics. One such mechanic is called “emancipation grills". When a player passes through an “emancipation grill", any previously created portals disappear. If a player was carrying a cube (or any other object) through an “emancipation grill", that cube or object is also destroyed.

Fig. 1 shows the fundamental technique used by Portal players to navigate the test chambers. A full list of game mechanics present can be found on the Portal Wiki. [3]

[1] https://store.steampowered.com/app/400/Portal/
[2] https://store.steampowered.com/app/620/Portal_2/
[3] https://theportalwiki.com/wiki/Category:Mechanics



(a) Fundamental technique to jump over chasms using portals (b) Building vertical height using portals

Figure 1: Demonstration of “flinging" – a fundamental technique to avoid and jump over obstacles in Portal and Portal 2. Image credit: `https://theportalwiki.com/wiki/Flinging`

3 Interesting Technical Aspects

This section briefly discusses novel and interesting technical aspects of applying reinforcement learning to Portal.

Unlike complex-to-understand real-time strategy games like DOTA 2 (Berner et al. (2019)) or StarCraft II (Vinyals et al. (2019)), Portal is conceptually much simpler, yet provides a basis to form incredibly difficult puzzles. In Portal or Portal 2, an expert player is NOT someone who knows the mechanics really well, but someone who can apply them in novel ways to get out of tricky situations. This makes even the most experienced players spend hours on a new puzzle they haven’t seen before.

Portal gameplay can also be compared with racing games such as Grand Turismo (Fuchs et al. (2021)) where the goal is to learn the car mechanics and learn to drive the car correctly along the racing track. As compared to these racing games, Portal is very difficult and different. In a car racing game, even a realistic one such as Gran Turismo, the driving mechanics can only be composed in ways that make sense in real life, in Portal, the game mechanics can be composed in any arbitrary way. This arbitrary composition is what makes Portal one of the best puzzle games of all time.

Becoming a good Portal player naturally requires the player to play a lot of different test chambers in order to learn common ways game mechanics are chained together. For example, vertical height gain using flinging is often combined with horizontal flinging to jump across a very long and tall chasm. Along with this, patterns such as “moving a cube around an emancipation grill using 3 portals" are common puzzle solving techniques.

Learning agents that can understand these simple mechanics and apply them in novel ways to always solve a test-chamber is an interesting technical challenge. This work proposes to present an RL environment that is inspired from Portal with the target task of “navigation".

4 Related Work

Building an environment for reinforcement learning can be a quite challenging task. Works that focus on retro gaming consoles such as Atari or Nintendo Bellemare et al. (2013); Nichol et al. (2018) use readymade emulators (e.g. Bellemare et al. (2013) used Stella emulator) to expose controls to the agent. First person shooter environment based on Doom game by Kempka et al. (2016) uses ZDoom [4] – an implementation Doom’s engine for the modern systems along with FreeDoom [5] providing the game assets and textures. DeepMind Lab (Beattie et al. (2016)) is another first-person 3D environment based on ioquake3[6] engine. Large scale works on DOTA 2 (Berner et al. (2019)) and StarCraft II (Vinyals et al. (2019)) used the game’s scripting API (DOTA 2 bot scripting API and StaCraft II Machine Learning API respectively) to control the agent and retrieve observations from the game directly. Owing to the popularity of games in reinforcement learning, game developers and researchers have also collaborated to form “general" game environment toolkits that can convert any game programmed in a specific game engine into a reinforcement learning environment e.g. Unity ML agents Juliani et al. (2020) and Godot reinforcement learning agents Beeching et al. (2021). Discussing performance benefits of various methods is out of scope of this document but is crucial for building final project.

[4] `https://zdoom.org`
[5] `https://freedoom.github.io`
[6] `https://ioquake3.org`

5 Scope and Execution Plan

The scope of this project is to only implement portal guns i.e. the agent should be able to shoot portals on walls and be able to move instantaneously between them. While many other mechanics are present in the game, the basic portal mechanics in itself is sufficient for creating very interesting and difficult puzzles. To stay reasonable, other mechanics are not being committed right now. This interesting mechanic can be implemented in multiple ways, the project aims to compare all of them and pick the one that works best in the end. Following are some of the ways portals can be implemented:

- As a portal gun mod for existing game engine that has been exposed for reinforcement learning e.g. ViZDoom (Kempka et al. (2016)) , DeepMind Lab (Beattie et al. (2016)) or Minecraft (Johnson et al. (2016))
- Implementing from scratch in a reinforcement learning enabled game engine e.g. Unity ML agents (Juliani et al. (2020)) or Godot reinforcement learning agents (Beeching et al. (2021))
- Implementing from scratch in a completely standalone environment
- Exposing Portal 2's Source engine for reinforcement learning

For each of these ways, at least 2 weeks are allotted for systematic investigation. Therefore, by April 1st, I expect that the core implementations of Portals will be The remaining time in the project will be used to polish the interface and training a demo agent using the developed interface. Any further extra time will be used to implement more functionality such as cubes, lasers and emancipation grills in the interface.

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